

bb-action-router

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Overview

Remote action router API

Use case: action level containers

Remote action router API

```
service ActionRouter {
  rpc RouteAction(RouteActionRequest) returns (RouteActionResponse);
}

message RouteActionRequest {
  string instance_name = 1;
  build.bazel.remote.execution.v2.DigestFunction.Value digest_function = 2;
  build.bazel.remote.execution.v2.Action action = 3;
}

message RouteActionResponse {
  build.bazel.remote.execution.v2.Action action = 1;
  repeated google.protobuf.Any invocation_keys = 4;
}
```

Source: https://github.com/buildbarn/bb-remote-execution/blob/main/pkg/proto/remoteactionrouter/remoteactionrouter.proto

Remote action router API

in_memory_build_queue.go / InMemoryBuildQueue.Execute

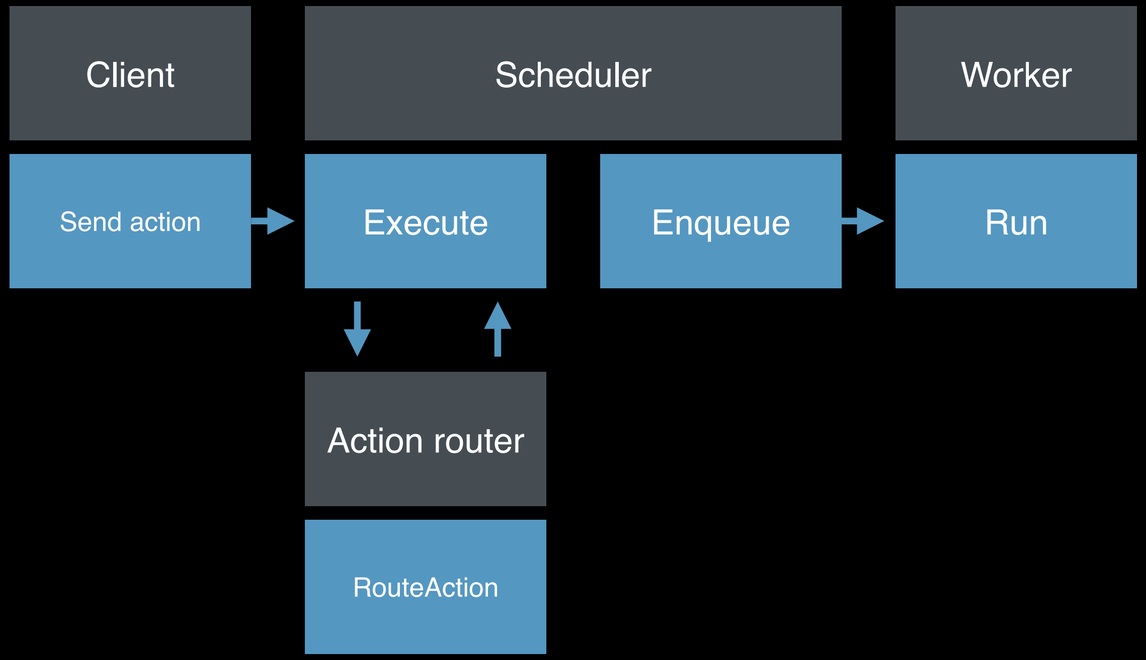
```
- platformKey, invocationKeys, initialSizeClassSelector, err :=
bq.actionRouter.RouteAction(ctx, actionDigest.GetDigestFunction(), action, requestMetadata)

+ action, platformKey, invocationKeys, initialSizeClassSelector, err :=
bq.actionRouter.RouteAction(ctx, actionDigest.GetDigestFunction(), action, requestMetadata)
```

Source: https://github.com/buildbarn/bb-remote-execution/pull/205/changes

Remote action router API

Action flow



Remote action router API

```
message Action {
  Digest command_digest = 1;
  Digest input_root_digest = 2;
  ...
}

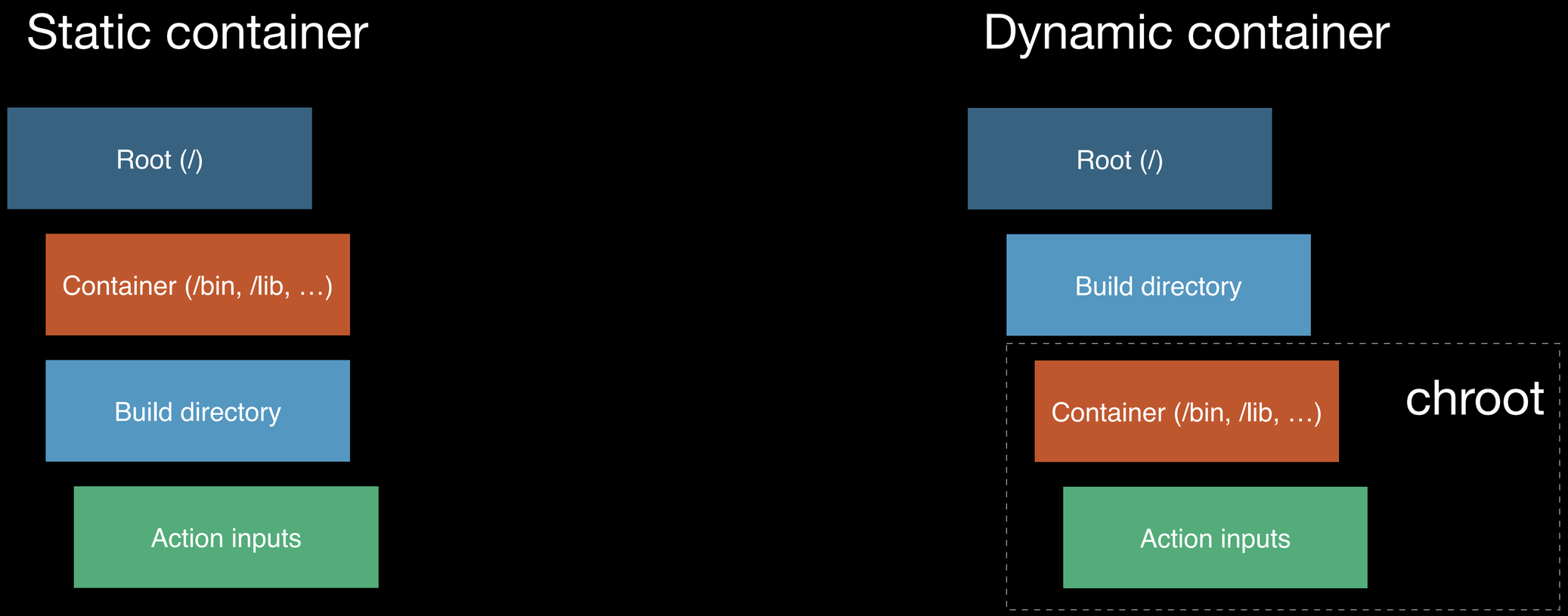
message Command {
  repeated string arguments = 1;.
  repeated EnvironmentVariable environment_variables = 2;
  repeated string output_files = 3;
  Platform platform = 5;
  string working_directory = 6;
}
```

Source: https://github.com/bazelbuild/remote-apis/blob/main/build/bazel/remote/execution/v2/remote_execution.proto

Use case: action level containers

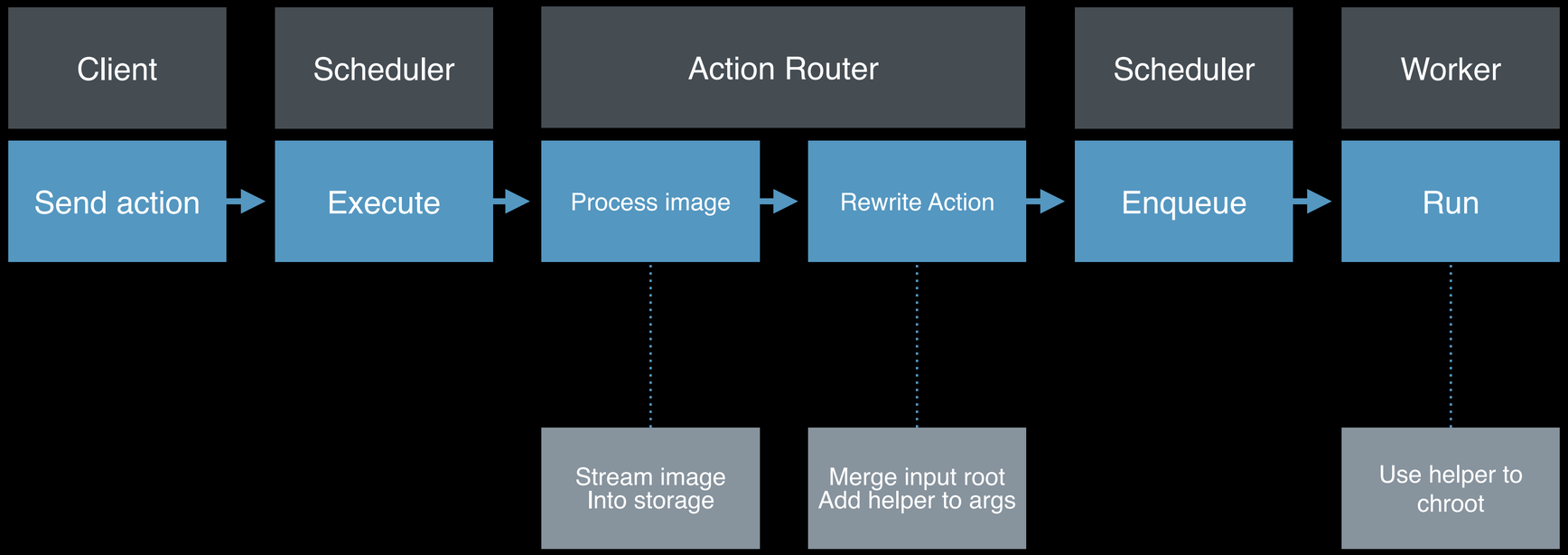
Action level containers

Static vs dynamic



Action level containers

Image merged with inputs flow



Action level containers

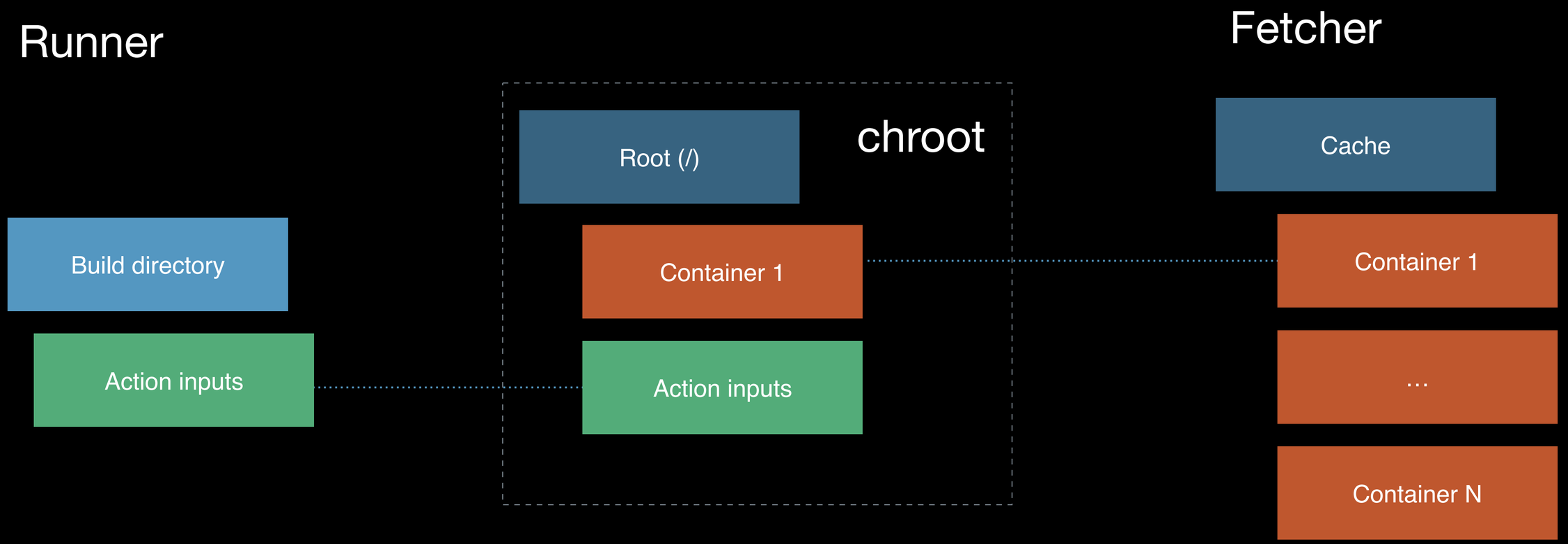
- Container filesystem becomes part of input tree
- Helper program performs chroot + exec of the original action

Action level containers

Speed

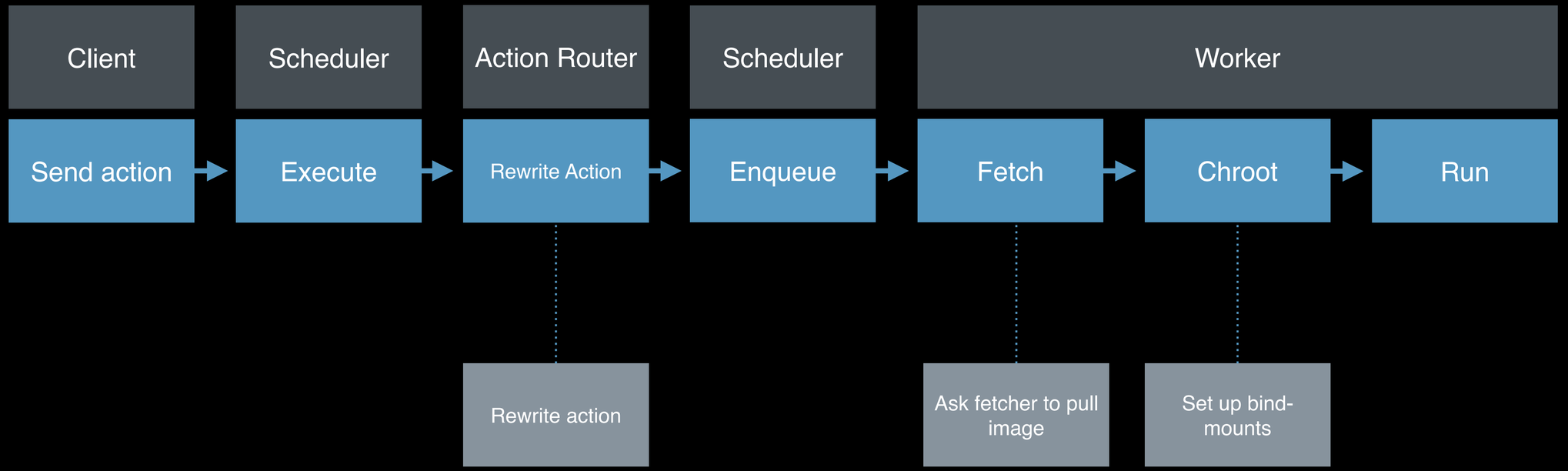
Action level containers

Container fetcher



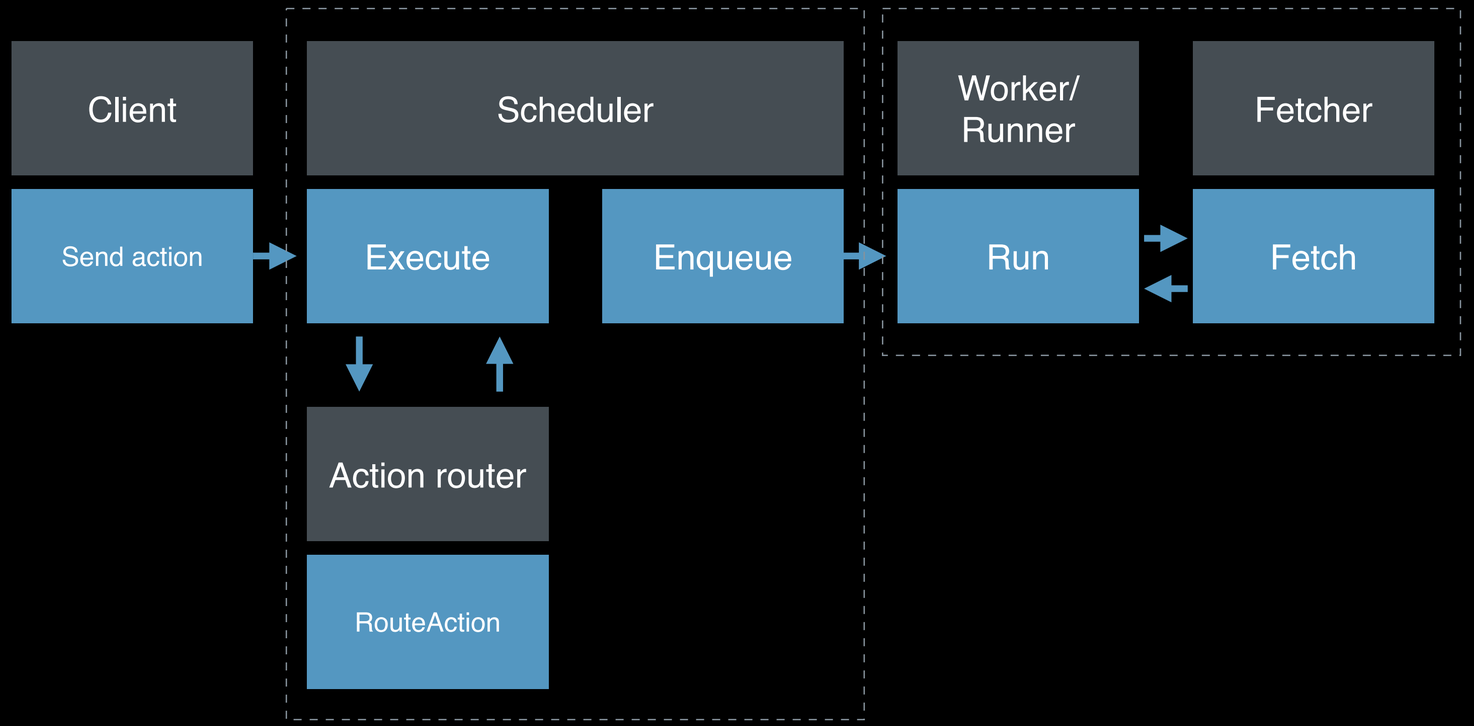
Action level containers

Image managed by fetcher flow



Remote action router API

Fetcher service services diagram



Action level containers

- Container filesystem managed in a separate service running on the worker
- Separate caches
- Separate caches Fetch becomes part of action execution
- Helper program is more complex
- Low overhead



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